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CS 621 DATA ANALYTICS

About Data

WE FLY

TO BE DISCUSSED

- Here, we give a brief overview on
 - Data domains
 - Data storages and formats
 - Structured data, Semi-structured data, Unstructured data
 - Data sources and examples



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DATA DOMAINS

- Domains rich with data:
 - Business domain
 - Scientific domain
 - Health Care domain
 - Web/Social Network domains
 - Manufactures
 - Others
- How data got collected?
 - Servers, devices, sensors, ... cloud.



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HOW DATA GOT STORED?

- Databases
- XML
- JSON
- TEXT files
 - csv, txt,
- Audio, Images, Videos
 - mp3, mp4, jpg, png,



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DATA TYPES, FORMATS

- Numerical Data
- Text Data
- Temporal Data
- Spatial Data
- Binary Data
- Multimedia Data
- Streaming Data
- Network Data

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UN-, SEMI-, STRUCTURED DATA

- Structured Data:
 - Stored in well-defined databases
 - Can be retrieved via standard languages, e.g., SQL
 - Mainly focus on the “read” operation.
 - Relational Database.

UN-, SEMI-, STRUCTURED DATA

- Semi-structured Data:
 - Stored in XML, JSON files
 - Again, read/load is our focus.
 - CSV files

UN-, SEMI-, STRUCTURED DATA

- Unstructured Data:
 - news articles, books, images, videos,
 - Need lots of cleaning, processing, shuffling, ... (rely on software programmers to do this job.)
 - Web pages, Email, server log files, ...

FREE DATA SOURCES AND EXAMPLES

- Many free online data repositories, for example

- Kdnuggets

<https://www.kdnuggets.com/datasets/index.html>

(raw; not cleaned)

- Kaggle

<https://www.kaggle.com/datasets>

(most of them are cleaned)



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DATA EXAMPLE (HOUSING DATA FROM KAGGLE)

longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value	ocean_proximity
-122.23	37.88	41	880	129	322	126	8.3252	452600	NEAR BAY
-122.22	37.86	21	7099	1106	2401	1138	8.3014	358500	NEAR BAY
-122.24	37.85	52	1467	190	496	177	7.2574	352100	NEAR BAY
-122.25	37.85	52	1274	235	558	219	5.6431	341300	NEAR BAY
-122.25	37.85	52	1627	280	565	259	3.8462	342200	NEAR BAY



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DATA EXAMPLE (FROM AMAZON)



John Chellan



Amazing Laptop

Reviewed in the United States on May 9, 2021

Capacity: 256GB | Color: Space Gray | **Verified Purchase**

Amazing battery life. Super fast processor.

Helpful

Report abuse



Miles S.



Smelly

Reviewed in the United States on May 9, 2021

Capacity: 256GB | Color: Silver | **Verified Purchase**

Product smelled like it was stored in a warehouse that was moist and musty. I'm glad I bought the extended warranty.

Helpful

Report abuse



Camille Padilla



Love, love, love it!

Reviewed in the United States on May 9, 2021

Capacity: 512GB | Color: Gold | **Verified Purchase**

The ease of set up and linking to all of my other apple products was fantastic. This is fast, responsive and beautiful. Love it!!!

Helpful

Report abuse



Buyeruvstuff



Buy Apple from Apple, seriously...

Reviewed in the United States on May 9, 2021

Capacity: 256GB | Color: Space Gray | **Verified Purchase**

Yay, spent a G on a new grey screen of death. A month after the return was received by amazing... still no refund. Spend the extra \$150 and get it from Apple. Duh?

Helpful

Report abuse

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